THERMODYNAMICS

1. Agar system ko gain hua → (+)
   " " " loss " → (−).

   eg.→ Heat given to system → +50 J
   Heat lost by " → −50 J

2. $\Delta U = Q + W$ → work
   ↓ Tijory ↳ Investment

3. $\Delta U = n C_V \Delta T$
   $Q = n C \Delta T$
   $W = \int P dV$ (REV) (where P may / may not be const.)
   $= P_{ext} \Delta V$ (IRR)

4. 1 atmL = 101.325 Nm
   ↓ (J)

5. $C_p - C_V = R$
   ↓ ↳ $\frac{R}{\gamma - 1}$
   $\frac{\gamma R}{\gamma - 1}$

   $\frac{C_p}{C_V} = \gamma$

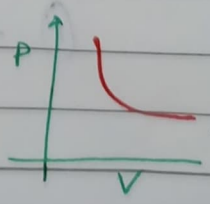
   $\gamma = 1 + \frac{2}{f}$ ($f \rightarrow 3, 5, 6$)

   $C_V = \frac{fR}{2}$ $C_p = \frac{\gamma f R}{2}$

6. Adiabatic → $Q = 0$ $PV^{\gamma} = C$

   $W = \frac{nR\Delta T}{\gamma - 1}$

   $C = C_V + \frac{R}{1 - \gamma}$



slope $= -\gamma \left(\frac{P}{V}\right)$

FLOT

	ISOBARIC (P const.)	ISOCHORIC (V const.)	ISOTHERMAL (T const.)	ADIABATIC ($\Delta Q = 0$)
$\Delta W \rightarrow$	$-P\Delta V$	Zero	$-nRT \ln \frac{V_2}{V_1}$ OR $-nRT \ln \frac{P_1}{P_2}$	$nC_V\Delta T \Rightarrow \frac{nR\Delta T}{\gamma - 1}$
$\Delta U \rightarrow$	$nC_V\Delta T$	$nC_V\Delta T$	Zero	$nC_V\Delta T$
$\Delta Q \rightarrow$	$nC_p\Delta T$	$nC_V\Delta T$	$+nRT \ln \frac{V_2}{V_1}$	Zero
$\Delta H \rightarrow$	$nC_p\Delta T$	$nC_p\Delta T$	Zero	$nC_p\Delta T$
FLOT $\rightarrow$	$\Delta U = \Delta W + \Delta Q$	$\Delta U = \Delta Q$	$\Delta W = -\Delta Q$	$\Delta U = \Delta W$
$C \rightarrow$	C_p	$C_T = \infty$	C_V	$C_A = 0$

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7. $\Delta H \rightarrow$ enthalpy change $= n C_p \Delta T$

8. Entropy

Spontaneous $\leftrightarrow$ increase in entropy.

$$\Delta S_{Total} = \Delta S_{sys} + \Delta S_{surrounding}$$

$$\boxed{dS = \frac{dQ_{REV}}{T}}$$

$$\rightarrow \quad \Delta S_{surr.} = -\frac{\Delta H_{sys}}{T} = -\frac{\Delta Q_{sys}}{T} \qquad \text{— (1)}$$

$$T(\Delta S_{Total}) = T(\Delta S_{sys}) - \Delta H_{sys} = -\Delta G$$

↗ Gibbs free Energy

$$\boxed{\Delta G = -T\Delta S_{total} = \Delta H_{sys} - T\Delta S_{sys}}$$

Process spont. $\Rightarrow$ $\Delta G < 0$ i.e $\Delta S_T > 0$

non spont. $\Rightarrow$ $\Delta G > 0$ i.e $\Delta S_T < 0$

REV $\Rightarrow$ $\Delta G = 0$ i.e $\Delta S_T = 0$

$T \uparrow \Rightarrow$ spontaneous $\uparrow$.

$$\Delta S = \int \frac{dQ_{REV}}{T} = \int \frac{dU + PdV}{T} = \int \frac{nC_V\, dT}{T} + \int \frac{PdV}{T}$$

$$\therefore \quad \Delta S_{sys} = nC_V \ln\frac{T_2}{T_1} + nR \ln\frac{V_2}{V_1} \qquad \text{— (2)}$$

$$(1) + (2) = \Delta S_T$$

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9. Adiabatic Rev aur IRR mein aap jagah bind nhi kar skte graph pe. Although ΔS is a state fnc, $\Delta S_{REV} \neq \Delta S_{IRR}$ because the final states in REV and IRR are not same!

Baaki saare processes mein $\Delta S_{REV} = \Delta S_{IRR}$ ✓

THERMOCHEMISTRY

1. Standard → 298K 1 atm

2. ΔH°_f → Compound 1 mole banega at standard cond^n.
   → Reactants must be in their reference states.

3. Reference states → C(s, graphite) P_4(white (yellow))
   Inn sabka $\Delta H^\circ_f = 0$ S_8(Rhombic) $H_2(g)$ $O_2(g)$ $N_2(g)$
   $F_2(g)$ $Cl_2(g)$ $Br_2(l)$ $I_2(s)$

4. $\Delta H^\circ_{combustion}$ → ① 1 mole burn karega
   ② std cond^n
   ③ Enough O_2

5. $\Delta H^\circ_{neutr.}$ → 1 eq. SA + 1 eq. SB → −13.7 kcal/mol
   −57.3 kJ/mol

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6. $\Delta H^\circ_{\text{atomization}}$ ⇒ ① $Cl_2 \rightarrow 2Cl$
② $Na_{(s)} \rightarrow Na_{(g)}$
③ $CH_4 \rightarrow C_{(g)} + 4H_{(g)}$

(gaseous atoms mein todna)

7. $\Delta H^\circ_{sol^n} = \underbrace{\Delta H^\circ_{lattice}}_{\text{Klagi}} + \underbrace{\Delta H^\circ_{hydration}}_{\text{Kmili}}$

8. $dH = dU + PdV$ ↖ at const P $\quad (H = U + PV)$

$\Delta H = \Delta U + \Delta n_g RT$

9. $G = H - TS$

$dG = VdP - SdT$

(Use $H = U + PV$, $dU = dQ + dW$, $dS = \frac{dQ}{T}$)

10. The entropy of a perfectly crystalline substance at 0K is 0.
{III Law} ↗

11. At const. T ⇒ $dG = VdP$

$\therefore \Delta G = nRT \ln \frac{P_2}{P_1}$

$\Delta G_{molar} = \Delta G^\circ_m + RT \ln P$